

A POSTERIORI ERROR ANALYSIS OF TWO STAGE COMPUTATION METHODS WITH APPLICATION TO EFFICIENT RESOURCE ALLOCATION AND THE PARAREAL ALGORITHM

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Abstract.

We consider the numerical solution of initial value problems that employ two stage computations consisting of a relatively coarse discretization followed by a relatively fine discretization. Motivation for such an approach includes adaptive error control, parallel-in-time solution schemes, and the numerical solution of adjoint problems for computing a posteriori error estimates. We describe a general formulation of two stage computations then perform a general a posteriori error analysis based on computable residuals and solution of an adjoint problem. The analysis accommodates various variations in the two stage computation and in formulation of the adjoint problems. We use the analysis to formulate novel procedures for the allocation of computational resources that take into account the cancellation of error. We also extend the analysis to derive a posteriori error estimates for the well-known Parareal Algorithm. We test the various results using several numerical examples.

Key words. a posteriori error estimation, adjoint problem, cancellation of error, efficient time step selection, initial value problem, parallel in time, Parareal algorithm, two stage computation

AMS subject classifications. 65N15, 68W10, 65N50, 65L05

1. Introduction. The general subject of this paper is a posteriori error analysis for the numerical solution of a system of ordinary differential equations,

$$\begin{cases} \dot{y} = f(y, t), & 0 < t < T, \\ y(0) = y_0. \end{cases} \quad (1.1)$$

The goal of the computation is to approximate the value of a quantity of interest (QoI) defined as,

$$Q(u) = \int_0^T (\psi(t), y(t)) dt, \quad (1.2)$$

for a specified $\psi(t) \in L_2(0, T]$, where $(\cdot, \cdot)$ is the Euclidean inner-product. Examples of ψ include $\psi \equiv 1/T$, giving the average, and approximate delta functions located at a time of interest.

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We consider standard discretizations of (1.1), e.g. finite difference and finite element methods arranged in solution approaches that employ two stages of computation in order to produce the final numerical solution;

1. A first stage solution computed on a relatively coarse discretization,
2. A second stage solution computed on a relatively fine discretization.

Two situations that employ two stage computations are:

Efficient allocation of computational resources. The first stage solution is used to compute information about the error. This information is used to generate a second stage discretization which achieves accuracy with some semblance of efficiency. Classically, this is performed by adaptive error control on a step by step basis. However, we investigate the possibility of exploiting cancellation of error by solving over relatively long time intervals in order to devise adaptive algorithms with improved efficiency.

Parallel-in-time methods. The two stages are used to devise algorithms that provide a way to exploit parallelization for time integration [27, 12, 19]. This generally involves iteratively exchanging information computed from coarse scale and fine scale numerical solutions at time nodes that span a number of numerical time steps used for each solution. We focus attention on the *Parareal Algorithm* [30, 28, 29, 31, 20, 21].

The primary focus of this paper is the derivation and testing of accurate a posteriori estimates for the error in the QoI computed from a two stage computation. One motivation is that such estimates are a critical component of uncertainty quantification in which all sources of error and uncertainty have to be quantified, e.g. see [17, 18]. This is especially important in situations in which adaptive error control or parallel-in-time solutions are employed, since those correspond to situations in which numerical error is expected to be significant. Accurate error estimation is also important for the purpose of answering issues related to efficiency of solution, e.g. determining the gain in accuracy obtained from a two stage computation and devising efficient algorithms for distributing computational resources.

A posteriori error estimation provides another motivation for the use of a two stage computation. We use a posteriori error analysis that involves computational residuals of the numerical solution, the solution of an adjoint problem for (1.1), and variational analysis to produce an *accurate* error estimate for the QoI [15, 14, 16, 3, 1]. The solution of the adjoint problem is critical to achieving accuracy as it provides a way to accurately quantify the effects of stability, e.g. cancellation and propagation of error. However, the numerical solution of the adjoint problem can involve significant computational cost, motivating the study:

Computation of a posteriori error estimates. Use a two stage computation to reduce the computational cost of computing numerical solutions of an adjoint equation.

In §2, we introduce an abstract two stage computation which models the typical paradigm of a numerical solution on a coarse scale followed by a computation at a finer scale. We derive various a posteriori error estimates for the general two stage computation in §3, allowing for variations in the forward solve and in the solution of an adjoint problem. We describe new strategies for distributing computational resources to exploit the accuracy of a posteriori error estimates in §4 and we extend the error estimates to the Parareal algorithm in §5. Conclusions are presented in §6.

2. Generic formulation of two stage computation. We begin by providing a generic formulation of a two stage computation. Stage 1 is a relatively coarse dis-

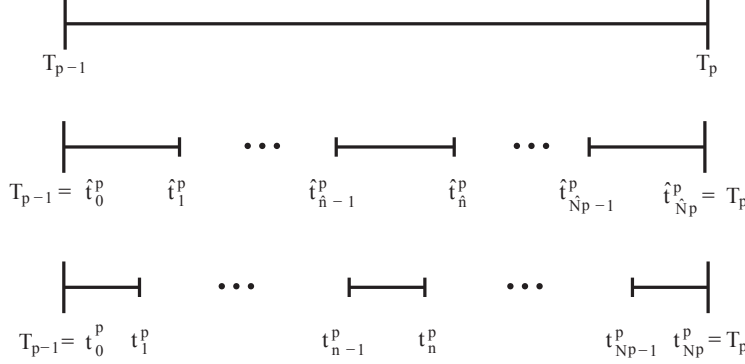


Fig. 2.1: Top: Subdomain $[T_{p-1}, T_p]$. Middle: Stage 1 discretization for $[T_{p-1}, T_p]$. Bottom: Stage 2 discretization for $[T_{p-1}, T_p]$.

cretization for which we assume it is possible to compute a classic a posteriori error estimate including solution of an adjoint problem. Stage 2 is a refined, relatively accurate discretization computed with various modifications of a standard discretization. We also consider various modifications in the standard computation of an a posteriori estimate for the Stage 2 solution.

Let $0 = T_0 < \dots < T_{p-1} < T_p < \dots < T_P = T$ be a partition of the time domain. The Stage 1 and Stage 2 solves may exchange information at these time nodes, which can be viewed as “synchronization” times. We call the time interval $[T_{p-1}, T_p]$ subdomain p . For the clarity of exposition, we consider the continuous Galerkin method for each stage. The analysis extends to a large variety of finite element and finite difference schemes. We comment on this briefly in §3.3.

2.1. Stage 1. We compute the approximation $\hat{Y}^p(t)$ of (1.1) in $(T_{p-1}, T_p]$ with initial condition α at $t = T_{p-1}$ using a numerical scheme $\hat{G}^p[\alpha](t)$, where the superscript p indicates that $t \in (T_{p-1}, T_p]$. We assume the numerical solver $\hat{G}$ is the continuous Galerkin method of order q , $cG(q)$. We partition $[T_{p-1}, T_p]$ as $T_{p-1} = \hat{t}_0^p < \dots < \hat{t}_{N^p-1}^p < \hat{t}_{N^p}^p = T_p$, as shown in Figure 2.1, with $\hat{I}_{\hat{n}}^p = [\hat{t}_{\hat{n}-1}^p, \hat{t}_{\hat{n}}^p]$. The approximation is in the space $\hat{\mathcal{V}}_{\hat{n}}^{p,q}$, which is the set of polynomials of degree q on the interval $\hat{I}_{\hat{n}}^p$. The $cG(q)$ method is to find $\hat{Y}^p \in \hat{\mathcal{V}}_{\hat{n}}^{p,q}$ such that,

$$\int_{\hat{t}_{\hat{n}-1}^p}^{\hat{t}_{\hat{n}}^p} (\dot{\hat{Y}}^p, v) dt = \int_{\hat{t}_{\hat{n}-1}^p}^{\hat{t}_{\hat{n}}^p} (f(\hat{Y}^p, t), v) dt, \quad \forall v \in \hat{\mathcal{V}}_{\hat{n}}^{p,q-1}, \quad (2.1)$$

for $1 \leq \hat{n} \leq \hat{N}^p$ and $\hat{Y}^p(T_{p-1}) = \hat{z}_{p-1}$. For $p = 1$, we set $\hat{Y}^p(0) = y(0)$. If we set $\hat{z}_{p-1} = \hat{Y}^{p-1}(T_{p-1})$, we obtain the global $cG(q)$ method on $[0, T]$, which we call the “serial” solution. Other options for $\hat{z}_{p-1}$ are explored in §5.

2.2. Stage 2. For Stage 2, we use a second numerical scheme $F^p[\alpha](t)$ which is more accurate than $\hat{G}$, e.g. employing a finer discretization or a higher order method. For simplicity in this paper, we use the $cG(q)$ method with a finer partition as $T_{p-1} = t_0^p < \dots < t_{N^p-1}^p < t_{N^p}^p = T_p$ as shown in Figure 2.1. We let $I_n^p = [t_{n-1}^p, t_n^p]$. The analysis easily handles other discretization choices. The $cG(q)$ method is to find

$Y^p \in \mathcal{V}_n^{q,p}$ such that,

$$\int_{t_{n-1}^p}^{t_n^p} (\dot{Y}^p v) dt = \int_{t_{n-1}^p}^{t_n^p} (f(Y^p, t), v) dt, \quad \forall v \in \mathcal{V}_n^{p,q-1}, \quad (2.2)$$

with initial conditions $Y^p(T_{p-1}) = z_{p-1}$. For subdomain 1, we always have $z_0 = y(0)$. For the other subdomains, we consider two options for conditions at the left-hand time node:

1. Set $z_{p-1} = Y^{p-1}(T_{p-1})$ yielding the standard cG(q) method on $[0, T]$.
2. Initialize the solution in each subdomain with the values from the solution from Stage 1 by setting $z_{p-1} = \hat{z}_{p-1} = \hat{Y}^p(T_{p-1})$. In this case, the solutions in each subdomain may be computed in parallel. Moreover, the solution is continuous on each subdomain p , but it may be discontinuous at time nodes T_p for $1 < p < P-1$.

REMARK 1. We use $\hat{\cdot}$ to indicate a Stage 1 quantity and a super-script p to reference the subdomain $[T_{p-1}, T_p]$.

3. A posteriori analysis for two stage computation. Our aim is to investigate strategies to approximate error in a quantity of interest for the two stage computation in an accurate yet computationally efficient manner. Furthermore, we need accurate error estimation techniques that apply to both serial and parallel-in-time approaches. There are multiple alternatives for the a posteriori analysis of the two stage computation, involving a trade-off between accuracy and computational cost through various choices for the adjoint problem.

3.1. A posteriori analysis for Stage 1. We use the traditional a posteriori error analysis for Stage 1. The error in the QoI is defined as,

$$\int_0^T (\hat{e}^p, \psi) dt = \int_0^T (y - \hat{Y}^p, \psi) dt.$$

The Stage 1 adjoint is,

$$\begin{cases} -\dot{\hat{\phi}}(t) = \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi} + \psi(T), & t \in [0, T], \\ \hat{\phi}(T) = 0, \end{cases} \quad (3.1)$$

where $\overline{f'(y, \hat{Y}^p)} = \int_0^1 \frac{\partial}{\partial s} f(sy + (1-s)\hat{Y}^p) ds$ is the standard linearization choice. This implies,

$$\overline{f'(y, \hat{Y}^p)}(y - \hat{Y}^p) = f(y) - f(\hat{Y}^p). \quad (3.2)$$

Note that the Stage 1 adjoint is continuous in $[0, T]$.

LEMMA 3.1.

On $[T_{p-1}, T_p]$, we have,

$$(\hat{\phi}, \hat{e}^p)(T_p) + \int_{T_{p-1}}^{T_p} (\psi, \hat{e}^p) dt = (\hat{\phi}, \hat{e}^p)(T_{p-1}) + \sum_{\hat{n}=1}^{\hat{N}^p} \int_{\hat{t}_{\hat{n}}^p} (\hat{\mathcal{R}}^p, \hat{\phi}) dt, \quad (3.3)$$

where $\hat{\mathcal{R}}^p = -\dot{\hat{Y}}^p + f(\hat{Y}^p)$ is the Stage 1 residual.

Proof.

From (3.1) and (3.2) on $\hat{I}_{\hat{n}}^p$,

$$\begin{aligned} \int_{\hat{I}_{\hat{n}}^p} (\psi, \hat{e}^p) dt &= \int_{\hat{I}_{\hat{n}}^p} (-\dot{\hat{\phi}} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, \hat{e}^p) dt \\ &= -(\hat{\phi}, \hat{e}^p)_{\hat{n}, p} + (\hat{\phi}, \hat{e}^p)_{\hat{n}-1, p} + \int_{\hat{I}_{\hat{n}}^p} (\hat{\phi}, \dot{y} - \dot{\hat{Y}}^p) - (\hat{\phi}, f(y) - f(\hat{Y}^p)) dt. \end{aligned}$$

Now (1.1) implies,

$$\int_{\hat{I}_{\hat{n}}^p} (\psi, \hat{e}^p) dt = -(\hat{\phi}, \hat{e}^p)_{\hat{n}, p} + (\hat{\phi}, \hat{e}^p)_{\hat{n}-1, p} + \int_{\hat{I}_{\hat{n}}^p} (\hat{\mathcal{R}}^p, \hat{\phi}) dt. \quad (3.4)$$

Summing (3.4) from $\hat{n} = 1$ to $\hat{n} = \hat{N}^p$ yields (3.3).

□

THEOREM 3.1. *The error in the QoI for the Stage 1 solution satisfies,*

$$\int_0^T (\psi, \hat{e}^p) dt = \sum_{p=2}^P (\hat{\phi}(T_{p-1}), \hat{Y}^{p-1}(T_{p-1}) - \hat{z}_{p-1}) + \sum_{p=1}^P \hat{D}^p(\hat{Y}^p, \hat{\phi}),$$

where $\hat{D}^p(\hat{Y}^p, \hat{\phi}) = \sum_{\hat{n}=1}^{\hat{N}^p} \int_{\hat{I}_{\hat{n}}^p} (\hat{\mathcal{R}}^p, \hat{\phi}) dt$ represents the error contribution on $[T_{p-1}, T_p]$.

Proof. Summing (3.3) for $p = 1, \dots, P$,

$$\int_0^T (\psi, \hat{e}^p) dt = \sum_{p=1}^P \left[-(\hat{\phi}, y - \hat{Y}^p)(T_p) + (\hat{\phi}, y - \hat{Y}^p)(T_{p-1}) + \hat{D}^p(\hat{Y}^p, \hat{\phi}) \right].$$

Since $\hat{\phi}$ and y are continuous in $[0, T]$, $y(0) = \hat{Y}^p(0)$ and $\hat{\phi}(T) = 0$ we have,

$$\begin{aligned} \int_0^T (\psi, \hat{e}^p) dt &= \sum_{p=1}^{P-1} (\hat{\phi}, \hat{Y}^p)(T_p) - \sum_{p=2}^P (\hat{\phi}, \hat{Y}^p)(T_{p-1}) + \sum_{p=1}^P \hat{D}^p(\hat{Y}^p, \hat{\phi}), \\ &= \sum_{p=2}^P (\hat{\phi}, \hat{Y}^{p-1} - \hat{Y}^p)(T_{p-1}) + \sum_{p=1}^P \hat{D}^p(\hat{Y}^p, \hat{\phi}). \end{aligned}$$

Finally observing that $\hat{Y}^p(T_{p-1}) = \hat{z}_{p-1}$ proves the result.

□

An immediate consequence follows.

COROLLARY 3.1. *Let $\hat{z}_{p-1} = \hat{Y}^{p-1}(T_{p-1})$, then,*

$$\int_0^T (\psi, \hat{e}^p) dt = \sum_{p=1}^P \hat{D}^p(\hat{Y}^p, \hat{\phi}). \quad (3.5)$$

REMARK 2. *These error representations use exact adjoint solutions defined by linearization employing an average between the true and numerical solutions. In practice, an adjoint problem is approximated by linearization around the numerical solution and the adjoint solutions are computed numerically. Hence, we obtain an approximate error estimate from the exact error representation. This is well established in the literature [16]. This approach is employed for all of the adjoint problems discussed in this paper. For the sake of simplicity, we do not write approximate versions of the exact error representations.*

3.2. A posteriori analysis for Stage 2. The error representation for Stage 2 depends on how the solutions Y^p on each subdomain p are initialized, as well as how we define the adjoint problem.

3.2.1. Analysis for Stage 2 using the Stage 1 adjoint problem. In this approach, we use the adjoint solution computed for Stage 1 to form an estimate. The different linearization points for the adjoint problems for Stage 1 and 2 means that the analog of (3.2) does not hold.

LEMMA 3.2. *We have,*

$$\begin{aligned}
(\hat{\phi}, y - Y^p)(T_p) + \int_{T_{p-1}}^{T_p} (\psi, e^p) dt &= (\hat{\phi}, y - Y^p)(T_{p-1}) + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \hat{\phi}) dt \\
&\quad + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, \hat{Y}^p - Y^p) dt \\
&\quad + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, y - \hat{Y}^p) dt,
\end{aligned} \tag{3.6}$$

where $\mathcal{R}^p = -\dot{Y}^p + f(Y^p)$ is the Stage 2 residual.

Proof. On each interval $I_n^p \in (T_{p-1}, T_p)$ we have,

$$\begin{aligned}
\int_{I_n^p} (\psi, e^p) dt &= \int_{I_n^p} (-\dot{\hat{\phi}} - \overline{f'(y, \hat{Y}^p)})^\top \hat{\phi}, e^p) dt \\
&= \int_{I_n^p} (-\dot{\hat{\phi}} - \overline{f'(y, Y^p)})^\top \hat{\phi}, e^p) dt + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, e^p) dt \\
&= -(\hat{\phi}, e^p)_{n,p} + (\hat{\phi}, e^p)_{n-1,p} \\
&\quad + \int_{I_n^p} (\mathcal{R}^p, \hat{\phi}) + (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, e^p) dt.
\end{aligned}$$

Summing from $n = 1$ to $n = N^p$,

$$\begin{aligned}
\int_{T_{p-1}}^{T_p} (\psi, e^p) dt &= (\hat{\phi}, y - Y^p)(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) \\
&\quad + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \hat{\phi}) dt + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, e^p) dt.
\end{aligned}$$

Adding and subtracting $\int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, \hat{Y}^p) dt$ leads to,

$$\begin{aligned}
\int_{T_{p-1}}^{T_p} (\psi, e^p) dt &= (\hat{\phi}, y - Y^p)(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \hat{\phi}) dt \\
&\quad + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, \hat{Y}^p - Y^p) dt \\
&\quad + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, y - \hat{Y}^p) dt.
\end{aligned}$$

□

THEOREM 3.2. *The error in QoI for the Stage 2 solution using the adjoint for Stage 1 (3.1) is,*

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\hat{\phi}, Y^{p-1} - Y^p)(T_{p-1}) + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \hat{\phi}) dt \right. \\ &\quad + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, \hat{Y}^p - Y^p) dt \\ &\quad \left. + \int_{I_n^p} (\overline{f'(y, Y^p)})^\top \hat{\phi} - \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}, y - \hat{Y}^p) dt \right]. \end{aligned} \quad (3.7)$$

Proof. The proof follows from Lemma 3.2.

□

The last term in the error representation (3.6) involves the true solution y . It is not computable but still may be large. We illustrate this in §3.4.

3.2.2. Analysis for Stage 2 using various Stage 2 adjoints. The Stage 1 adjoint problem leads to a non-computable term in the resulting error representation. We thus seek to use an adjoint problem obtained by linearization involving the Stage 2 solution, but with various modifications aimed to yield a computationally inexpensive adjoint problem. The Stage 2 adjoint on the p^{th} subdomain is defined as,

$$\begin{cases} -\dot{\phi}^p(t) = \overline{f'(y, Y^p)}^\top \phi^p + \psi(T), & t \in [T_{p-1}, T_p], \\ \phi^p(T_p) = w_p. \end{cases} \quad (3.8)$$

Different choices of the initial condition w_p lead to different estimates and solution strategies. We always have $w_P = 0$. We describe and analyze three strategies for $w_p, p = 1, \dots, P-1$.

1. A globally continuous (i.e. continuous in $[0, T]$) adjoint solution resulting with choice $w_p = \phi^{p+1}(T_p)$.
2. Choosing $w_p \neq \phi^{p+1}(T_p)$, e.g. set $w_p = 0, p = 1, \dots, P-1$. This yields an adjoint solution that is continuous in a subdomain p but may be discontinuous across $T_p, p = 1, \dots, P-1$. This opens the possibility of solving the subdomain adjoint problems in parallel.
3. Set the initial conditions for Stage 2 adjoint using the Stage 1 adjoint, $w_p = \hat{\phi}(T_p)$.

Now we derive a generic result. On each interval $I_n^p = (T_{p-1}, T_p)$ we have,

$$\begin{aligned} \int_{I_n^p} (\psi, e^p) dt &= \int_{I_n^p} (-\dot{\phi}^p - \overline{f'(y, Y^p)}^\top \phi^p, e^p) dt \\ &= -(\phi^p, e^p)_{n,p} + (\phi^p, e^p)_{n-1,p} + \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned}$$

Summing from $n = 1$ to $n = N^p$, the continuity of y and $Y^p(T_{p-1}) = z_{p-1}$ and $\phi^p(T_p) = w_p$, we have,

$$\int_{T_{p-1}}^{T_p} (\psi, e^p) dt = (\phi^p, y - z_{p-1})(T_{p-1}) - (w_p, y - Y^p)(T_p) + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.9)$$

Summing (3.9) over all P subdomains,

$$\begin{aligned} & \int_0^T (\psi, e^p) dt \\ &= \sum_{p=1}^P (\phi^p, y - z_{p-1})(T_{p-1}) - (w_p, y - Y^p)(T_p) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned} \quad (3.10)$$

Now we investigate different choices for z_{p-1} and w_p .

Analysis using the global Stage 2 adjoint problem. We assume the initial conditions, $w_p = \phi^{p+1}(T_p)$, which leads to a globally defined adjoint as in the classic analysis. From (3.10) we have,

$$\begin{aligned} & \int_{T_{p-1}}^{T_p} (\psi, e^p) dt \\ &= \sum_{p=1}^P (\phi^p, y - z_{p-1})(T_{p-1}) - (\phi^{p+1}, y - Y^p)(T_p) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned} \quad (3.11)$$

We consider the two choices, (i) $z_{p-1} = Y^{p-1}(T_{p-1})$ and (ii) $z_{p-1} = \hat{Y}^p(T_{p-1})$.

THEOREM 3.3. *Let $z_{p-1} = Y^{p-1}(T_{p-1})$ for $p = 2, \dots, P$ and $w_p = \phi^{p+1}(T_p)$ for $p = 1, \dots, P-1$. Then,*

$$\int_0^T (\psi, e^p) dt = \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.12)$$

Proof. The proof follows directly from (3.11). $\square$

THEOREM 3.4.

Let $z_{p-1} = \hat{Y}^p(T_{p-1})$ for $p = 2, \dots, P$ and $w_p = \phi^{p+1}(T_p)$ for $p = 1, \dots, P-1$. Then,

$$\int_0^T (\psi, e^p) dt = \sum_{p=2}^P (\phi^p, Y^{p-1} - \hat{Y}^p)(T_{p-1}) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.13)$$

A drawback to this error representation is that it requires a fine-scale serial solution of the adjoint problem. We discuss ways to amend this next.

Analysis using the Stage 2 adjoint problem with homogeneous initial conditions. In this strategy we set $w_p = 0$ in (3.10),

$$\int_0^T (\psi, e^p) dt = \sum_{p=2}^P (\phi^p, y - z_{p-1})(T_{p-1}) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.14)$$

We again consider the two choices, (i) $z_{p-1} = Y^{p-1}(T_{p-1})$ and (ii) $z_{p-1} = \hat{Y}^p(T_{p-1})$.

THEOREM 3.5.

Let $w_p = 0$ for $p = 1, \dots, P-1$ and $z_{p-1} = Y^{p-1}(T_{p-1})$ for $p = 2, \dots, P$. Then,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\phi^p, \hat{Y}^{p-1} - Y^{p-1})(T_{p-1}) + \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt \right] \\ &\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt, \end{aligned}$$

where each $\hat{\phi}^{p,a}$ solves the auxiliary “adjoint” problem,

$$-\dot{\hat{\phi}}^{p,a} = \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}^{p,a}, \quad t \in [0, T_{p-1}), \quad (3.15)$$

with initial conditions $\hat{\phi}^{p,a}(T_{p-1}) = \phi^p(T_{p-1})$.

Proof. From (3.14) we have,

$$\int_0^T (\psi, e^p) dt = \sum_{p=2}^P (\phi^p, y - Y^{p-1})(T_{p-1}) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.16)$$

Adding and subtracting appropriate terms leads to,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\phi^p, y - \hat{Y}^{p-1})(T_{p-1}) + (\phi^p, \hat{Y}^{p-1} - Y^{p-1})(T_{p-1}) \right] \\ &\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned} \quad (3.17)$$

The term $\sum_{p=2}^P (\phi^p, y - \hat{Y}^{p-1})(T_{p-1})$ involves the true solution y so is not computable. However, since this term involves the Stage 1 solution $\hat{Y}^{p-1}$, we can estimate this term by defining auxiliary adjoint problems. Consider the auxiliary QoI,

$$\Psi(y) = (\phi^p, y)(T_{p-1}). \quad (3.18)$$

Using the auxiliary adjoint in (3.15), we obtain,

$$(\phi^p, y - \hat{Y}^{p-1})(T_{p-1}) = \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt. \quad (3.19)$$

To compute all the terms in $\sum_{p=2}^P (\phi^p(T_{p-1}), y(T_{p-1}) - \hat{Y}^{p-1}(T_{p-1}))$, (3.19) is solved for $2 < p < P$ for $P-1$ corresponding auxiliary QoI's (3.18).

Combining (3.17) and (3.19) proves the lemma. $\square$

THEOREM 3.6.

Let $w_p = 0$ for $p = 1, \dots, P-1$ and $z_{p-1} = \hat{Y}^p(T_{p-1})$ for $p = 2, \dots, P$. Then,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\phi^p, \hat{Y}^{p-1} - \hat{Y}^p)(T_{p-1}) + \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt \right] \\ &\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}, \phi^p) dt. \end{aligned}$$

Proof. From (3.14) we have,

$$\begin{aligned}
\int_0^T (\psi, e^p) dt &= \sum_{p=2}^P (\phi^p, y - \hat{Y}^p)(T_{p-1}) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt \\
&= \sum_{p=2}^P \left[(\phi^p, y - \hat{Y}^{p-1})(T_{p-1}) + (\phi^p, \hat{Y}^{p-1} - \hat{Y}^p)(T_{p-1}) \right] \\
&\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt.
\end{aligned} \tag{3.20}$$

Treating the term $\sum_{p=2}^P (\phi^p, y - \hat{Y}^{p-1})(T_{p-1})$ as before proves the result. $\square$

Analysis using the Stage 2 adjoint problem initialized using the Stage 1 adjoint solution. In this strategy we set $w_p = \hat{\phi}(T_p)$ in (3.10),

$$\int_0^T (\psi, e^p) dt = \sum_{p=1}^P (\phi^p, y - z_{p-1})(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \tag{3.21}$$

We again consider the two choices, (i) $z_{p-1} = Y^{p-1}(T_{p-1})$ and (ii) $z_{p-1} = \hat{Y}^p(T_{p-1})$.

THEOREM 3.7. *Let $w_p = \hat{\phi}(T_p)$ for $p = 1, \dots, P-1$ and $z_{p-1} = Y^{p-1}(T_{p-1})$ for $p = 2, \dots, P$. Then,*

$$\begin{aligned}
\int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\phi^p - \hat{\phi}, \hat{Y}^{p-1} - Y^{p-1})(T_{p-1}) + \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt \right] \\
&\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt.
\end{aligned} \tag{3.22}$$

where each $\hat{\phi}^{p,a}$ solves an auxiliary adjoint problem,

$$-\dot{\hat{\phi}}^{p,a} = \overline{f'(y, \hat{Y}^p)}^\top \hat{\phi}^{p,a}, \quad t \in [0, T_{p-1}), \tag{3.23}$$

with initial conditions $\hat{\phi}^{p,a}(T_{p-1}) = \phi^p(T_{p-1}) - \hat{\phi}(T_{p-1})$.

Proof. We set $z_{p-1} = Y^{p-1}(T_{p-1})$ in (3.21) and use initial conditions to arrive at,

$$\begin{aligned}
\int_0^T (\psi, e^p) dt &= \sum_{p=1}^P \left[(\phi^p, y - Y^{p-1})(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) + \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt \right], \\
&= \sum_{p=2}^P (\phi^p - \hat{\phi}, y - Y^{p-1})(T_{p-1}) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt.
\end{aligned} \tag{3.24}$$

Adding and subtracting appropriate terms leads to,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P (\phi^p - \hat{\phi}, y - \hat{Y}^{p-1})(T_{p-1}) + \sum_{p=2}^P (\phi^p - \hat{\phi}, \hat{Y}^{p-1} - Y^{p-1})(T_{p-1}) \\ &\quad + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned} \quad (3.25)$$

The term $\sum_{p=2}^P (\phi^p - \hat{\phi}, y - \hat{Y}^{p-1})(T_{p-1})$ is not computable. We define $P-1$ auxiliary QoIs as,

$$\Psi(y) = (\phi^p - \hat{\phi}, y)(T_{p-1}).$$

Employing the adjoint (3.23) gives the following error representation,

$$(\phi^p - \hat{\phi}, y - \hat{Y}^{p-1})(T_{p-1}) = \sum_{k=1}^{p-1} \sum_{n=1}^{\hat{N}^k} \int_{\hat{t}_{n-1}^k}^{\hat{t}_n^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt. \quad (3.26)$$

Combining (3.25) and (3.26) proves the result $\square$

THEOREM 3.8.

Let $w_p = \hat{\phi}(T_p)$ for $p = 1, \dots, P-1$ and $z_{p-1} = \hat{Y}^p(T_{p-1})$ for $p = 2, \dots, P$. Then,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\hat{\phi}, Y^{p-1} - \hat{Y}^p)(T_{p-1}) + \sum_{k=1}^{p-1} \sum_{n=1}^{\hat{N}^k} \int_{\hat{t}_{n-1}^k}^{\hat{t}_n^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt \right. \\ &\quad \left. + (\phi^p - \hat{\phi}, \hat{Y}^{p-1} - \hat{Y}^p)(T_{p-1}) \right] + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \end{aligned} \quad (3.27)$$

Proof. We set $z_{p-1} = \hat{Y}^p(T_{p-1})$ in (3.21) and get,

$$\int_0^T (\psi, e^p) dt = \sum_{p=1}^P (\phi^p, y - \hat{Y}^p)(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) + \sum_{p=1}^P \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p) dt. \quad (3.28)$$

Considering the first sum in (3.28), applying boundary conditions and adding and subtracting appropriate terms, we arrive at,

$$\begin{aligned} &\sum_{p=1}^P \left[(\phi^p, y - \hat{Y}^p)(T_{p-1}) - (\hat{\phi}, y - Y^p)(T_p) \right] \\ &= \sum_{p=2}^P \left[(\phi^p - \hat{\phi}, y - \hat{Y}^p) + (\hat{\phi}, Y^{p-1} - \hat{Y}^p) \right] (T_{p-1}) \\ &= \sum_{p=2}^P \left[(\phi^p - \hat{\phi}, y - \hat{Y}^{p-1}) + (\phi^p - \hat{\phi}, \hat{Y}^{p-1} - \hat{Y}^p) + (\hat{\phi}, Y^{p-1} - \hat{Y}^p) \right] (T_{p-1}). \end{aligned} \quad (3.29)$$

The term $\sum_{p=2}^P (\phi^p - \hat{\phi}, y - \hat{Y}^{p-1})(T_{p-1})$ is treated as above to arrive at the representation (3.26). Combining (3.28), (3.29) and (3.26) completes the proof.

□

Theorems 3.5–3.8 involving solving a number of auxiliary adjoint problems. However, these auxiliary adjoint problems are solved on the Stage 1 discretization which is much coarser than the Stage 2 discretization, and are solved independently in parallel. Hence, this approach is computationally feasible.

3.3. Extension to other methods. The analysis easily extends to other general solution methods besides the cG(q) method. For example, this includes the implicit Euler, Crank-Nicolson, IMEX schemes, discontinuous Galerkin methods etc [5, 6, 4, 24, 8]. The analysis for Crank-Nicolson is the same, as it is equivalent to the cG(1) method. For other methods, the requisite changes are in the definitions of terms which involve the residuals, that is, terms involving $\hat{\mathcal{R}}^p$ and $\mathcal{R}^p$. To simplify notation, we rewrite (3.27) as,

$$\begin{aligned} \int_0^T (\psi, e^p) dt &= \sum_{p=2}^P \left[(\hat{\phi}(T_{p-1}), Y^{p-1}(T_{p-1}) - \hat{z}_{p-1}) + \mathcal{A}(\mathcal{R}^p, \hat{\phi}^{p,a}) dt \right. \\ &\quad \left. + (\phi^p(T_{p-1}) - \hat{\phi}(T_{p-1}), \hat{Y}^{p-1}(T_{p-1}) - \hat{z}_{p-1}) \right] + \sum_{p=1}^P \mathcal{D}(Y^p, \phi^p) dt, \end{aligned} \quad (3.30)$$

where $\mathcal{D}(Y^p, \phi^p) = \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^p, \phi^p)$ and $\mathcal{A}(\mathcal{R}^p, \hat{\phi}^{p,a}) = \sum_{k=1}^{p-1} \sum_{n=1}^{\hat{N}^k} \int_{\hat{t}_{n-1}^k}^{\hat{t}_n^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a})$. In fact, (3.30) now also gives the error representation other methods besides the cG(q) method by appropriate definitions of $\mathcal{D}(Y^p, \phi^p)$ and $\mathcal{A}(\mathcal{R}^p, \hat{\phi}^{p,a})$.

As an example, we consider the changes in $\mathcal{D}(Y^p, \phi^p)$ and $\mathcal{A}(\mathcal{R}^p, \hat{\phi}^{p,a})$ for the dG(q) methods. The dG(q) for Stage 1 is to find $\hat{Y}^p \in \hat{\mathcal{V}}_{\hat{n}}^{p,q}$ such that,

$$([\hat{Y}^p]_{n-1,p}, v^+) + \int_{\hat{t}_{n-1}^p}^{\hat{t}_n^p} (\dot{\hat{Y}}^p, v) dt = \int_{\hat{t}_{n-1}^p}^{\hat{t}_n^p} (f(\hat{Y}^p, t), v) dt, \quad \forall v \in \hat{\mathcal{V}}_{\hat{n}}^{p,q}, \quad (3.31)$$

for $1 \leq \hat{n} \leq \hat{N}^p$. Here $[U]_{n,p}$ denotes the jump across $t_{n,p}$ by $[U]_{n,p} = U_{n,p}^+ - U_{n,p}^-$, where $U_{n,p}^\pm = \lim_{t \rightarrow t_{n,p}^\pm} U(t)$. The changes in the error representation are given by the following formulas,

$$\mathcal{D}(Y^p, \phi^p) = \sum_{n=1}^{N^p} -([\dot{Y}^p]_{n-1,p}, \phi_{n-1,p}^p) + \int_{I_n^p} (-\dot{Y}^p + f(Y^p), \phi^p) dt,$$

and

$$\mathcal{A}(\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) = \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} -([\dot{\hat{Y}}^p]_{\hat{n}-1,p}, \hat{\phi}^{p,a}) + \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (-\dot{\hat{Y}}^p + f(\hat{Y}^p), \hat{\phi}^{p,a}).$$

The implicit Euler method is equivalent to the dG(0) method.

3.4. A numerical example. In this example, we estimate error in the Stage 2 cG(1) approximation, Y^p , for the nonlinear ODE,

$$\begin{cases} \dot{y} = -0.25 - \sin(\pi t)y^2, & 0 < t < T, \\ y(0) = 1.0. \end{cases} \quad (3.32)$$

We present results for four different estimates. In the first estimate, “Stage 1 Only”, we estimate the error using (3.6) by dropping the non-computable terms $\int_{I_n^p} \overline{(f'(y, Y^p))^\top \hat{\phi} - f'(y, \hat{Y}^p)^\top \hat{\phi}, y - \hat{Y}^p} dt$. The second estimate (3.12), “Stage 2 Classical”, is obtained by solving a global Stage 2 adjoint problem. The third estimate (3.22), “Stage 2 Parallel”, is based on computing the Stage 2 adjoint in parallel on each of the p subdomains initialized with Stage 1 adjoint solution values. Finally, we also have a fourth estimate, “Stage 2 Parallel No Auxiliary Terms”, based on dropping the auxiliary terms, $\sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{t_{\hat{n}-1}^k}^{t_n^k} (\hat{\mathcal{R}}^p, \hat{\phi}^{p,a}) dt$, in (3.22).

For the numerical solutions, we choose $T = 3$, the Stage 1 time step $\Delta\hat{t} = 0.1$ and the Stage 2 time step $\Delta t = 0.00625$. The adjoint solutions are numerically estimated by the cG(1) method with time steps that are four times smaller than the time steps for the Stage 1 and Stage 2 solutions. The QoI is specified by choosing $\psi(t) = 1$. The results are shown in Table 3.1, in which we also present the effectivity ratios,

$$\text{Effectivity ratio} = \frac{\text{Estimated error}}{\text{True error}}.$$

The results indicate that the “Stage 1 Only” estimate is quite inaccurate. The schemes “Stage 2 Classical” and “Stage 2 Parallel” are quite accurate, as expected. The scheme “Stage 2 Parallel No Auxiliary Terms” is not as accurate as the last two schemes, but is still reasonably accurate with an effectivity ratio of 1.09.

Estimator	Estimated Error	Effectivity Ratio
Stage 1 Only	4.12E-06	2.16
Stage 2 Classical	1.89E-06	0.99
Stage 2 Parallel	1.89 E-06	0.99
Stage 2 Parallel No Auxiliary Terms	2.08E-06	1.09

Table 3.1: Comparison of error estimates obtained by different strategies using the cG(1) method for the primal solutions. The true error is 1.90E-06.

We further investigate the effect of the auxiliary terms by using the implicit Euler method for the primal solutions $\hat{Y}^p$ and Y^p , and keeping the other parameters the same. The results are shown in Table 3.2, where we now see that “Stage 2 Parallel No Auxiliary Term” is quite inaccurate when compared to the “Stage 2 Parallel” estimator which still yields a good error estimate.

Estimator	Estimated Error	Effectivity Ratio
Stage 2 Parallel	-0.0015	1.07
Stage 2 Parallel No Auxiliary Terms	-0.0031	2.21

Table 3.2: Comparison of error estimates obtained by different strategies using the implicit Euler method for the primal solutions. The true error is -0.0014.

4. Efficient allocation of computational resources. In this section, we discuss strategies for efficient allocation of computational resources for the Stage 2 so-

lution, based on the accurate a posteriori error estimate computed for the Stage 1 solution.

An important observation is that computing *accurate* error estimates generally requires accounting for the effects of cancellation of error. This is especially important in regards to adaptive time stepping because standard adaptive strategies are based directly or indirectly on an error bound instead of an estimate. In particular, standard error analyses of the common finite difference and finite element methods generally yield an expression of the form [22, 23, 13, 11, 10, 9, 7, 25, 26, 32],

$$\text{error} \approx \sum_{n=1}^N \text{local error contributions}_n, \quad (4.1)$$

where N is the number of time steps and the form of the local contributions depends on the analysis and method. In standard developments of adaptive error control strategies, this is replaced by the bound,

$$|\text{error}| \leq \sum_{n=1}^N |\text{local error contributions}_n|, \quad (4.2)$$

or some variation obtained with additional estimation of the local contributions. The significance in (4.2) over (4.1) lies in the fact that the absolute local contributions always add a positive contribution to the error bound. The developments of adaptive strategies then introduce a model of computational “work” per step and the order of the method to obtain an optimization problem for the optimal sequence of time steps. The solution to the optimization problem leads to the conclusion that the absolute local error contributions should be equal. In some literature, this called the *Principle of Equidistribution*.

Such strategies are generally described as optimally efficient. However, this refers to control of the error bound (4.2) not to the control of the actual error (4.1). In the common situation in which the signs of the local contributions vary, so that there is significant cancellation among the local contributions, the error bound (4.2) can be significantly larger than the error estimate (4.1). In this case, adaptive time step control based on an error bound may have little relation to controlling the actual error. This is reflected in the general difficulty in choosing a suitable local error tolerance for a standard adaptive error control scheme in order to achieve a desired accuracy in the result.

To illustrate, we solve (3.32) using the implicit Euler method with a time step of 0.01 on $[0, 3]$. The QoI is taken to be the weighted average value of the solution by setting $\psi = 1$ in (1.2) and the error estimate for the QoI is -0.0023 . The local error contributions obtained from the Stage 1 a posteriori estimate are plotted in Figure 4.1. The figure shows a period of positive contributions followed by a period of negative contributions and finally another period of positive contributions. This leads to significant cancellation of error contributions over the entire time period of solution. If we refine intervals 11 to 20, corresponding to the time period $[0.1, 0.2]$, by a factor of 10, then the total error increases in magnitude to -0.0034 ! The reason is that the mesh refinement leads to a significant decrease in size of the positive error contributions, leading to a reduced cancellation of contributions overall.

To describe the cancellation of error contributions, we define the accumulated

contributions to the error at end of interval k as,

$$\text{Accumulated contributions}_k = \left| \sum_{n=1}^k \text{local error contributions}_n \right|, \quad (4.3)$$

for $k = 1, \dots, N$. The accumulated contributions corresponding to Figure 4.1 are plotted in Figure 4.1. This illustrates the effects of cancellation of error contributions.

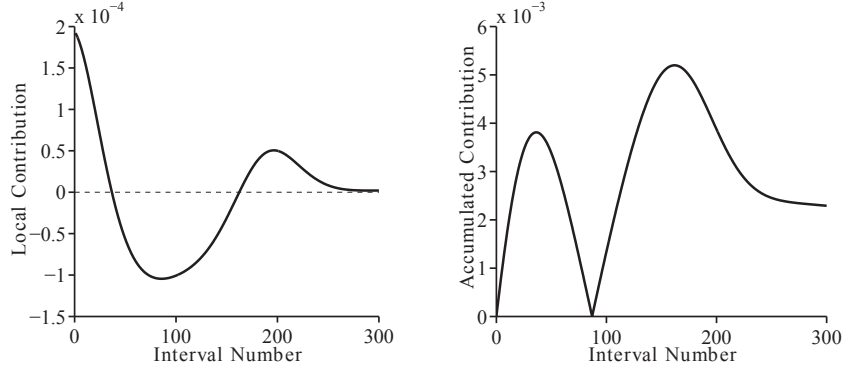


Fig. 4.1: Left: Local Error Contributions. Right: Accumulated Contributions to the Error. Note how the positive and negative contributions cancel each other to give the total error.

We indicated that *localized* time step refinement in a region over which significant cancellation of local error contributions occurs can lead to an increase in error. This appears to be a generic phenomena. Experimentation also suggests that employing *uniform* refinement over a region with significant cancellation of local error contributions preserves cancellation. For this reason, we propose an adaptive strategy that employs uniform refinement over time periods. The problem becomes identifying suitable time periods over which significant cancellation of local error contributions occur. We call such time periods “mesoscale regions” since they have size significantly larger than individual time steps but potentially much smaller than the entire time period of solution.

We assume that the Stage 1 solution is solved serially, that is, the Stage 1 solution in subdomain p gets its initial condition from the Stage 1 solution at the end of subdomain $p - 1$. For simplicity, we relabel $\hat{N} = \sum_{p=1}^P \hat{N}^p$, $\hat{I}_k = [\hat{t}_{k-1}, \hat{t}_k] = \hat{I}_n^p$ if $\hat{t}_{k-1} = \hat{t}_{n-1}^p$ and $\hat{t}_k = \hat{t}_n^p$ for some $\hat{n}$ and p , $\hat{Y}(t) = \hat{Y}^p(t)$ if $t \in [T_{p-1}, T_p]$ as illustrated in Figure 4.2.

We rewrite the Stage 1 error representation (3.5) as,

$$\int_0^T (\psi, \hat{e}) dt = \sum_{\hat{k}=1}^{\hat{N}} \int_{\hat{I}_{\hat{k}}} (-\dot{\hat{Y}} + f(\hat{Y}), \hat{\phi}) dt, \quad (4.4)$$

where $\hat{e} = y - \hat{Y}$, as illustrated in Figure 4.2. That is, we have now expressed (4.4) as the standard error representation for a global cG(q) solution. We define the

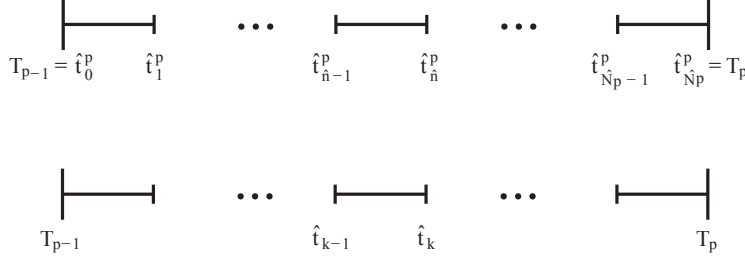


Fig. 4.2: Upper: Labeling used in §2. Lower: Relabeling for §4.

contribution of the interval $\hat{I}_{\hat{k}}$ to the error as $\hat{E}_{\hat{k}} = \int_{\hat{I}_{\hat{k}}} (-\hat{Y} + f(\hat{Y}), \hat{\phi}) dt$. Then (4.4) reads,

$$\int_0^T (\psi, \hat{e}) dt = \sum_{\hat{k}=1}^{\hat{N}} \hat{E}_{\hat{k}}. \quad (4.5)$$

4.1. A “mesoscale region” approach to grouping time steps. We define mesoscale regions as time intervals of “maximal” cancellation of local error contributions. We perform uniform refinement within each mesoscale region in order to preserve cancellation. We then seek to balance the contributions of the different mesoscale regions in order to obtain the desired accuracy in the error in the QoI.

Using a computed a posteriori error estimate, we compute accumulated contributions to the error as,

$$A\hat{E}_{\hat{k}} = \left| \sum_{\hat{i}=1}^{\hat{k}} \hat{E}_{\hat{i}} \right|. \quad (4.6)$$

This is easily obtained after the estimate has been computed.

We illustrate the accumulated contributions for the harmonic oscillator,

$$\frac{d^2 w}{dt^2} = -\frac{k}{m}w - \frac{c}{m} \frac{dw}{dt} + \frac{F_0}{m} \cos(\gamma t + \theta_d), \quad (4.7)$$

with $k = 50, m = 0.25, c = 1, F_0 = 50, \theta_d = 0, \gamma = 10$. The solution is shown in Figure 4.3. The second-order ODE is converted to a first-order ODE by setting $y_1 = w, y_2 = \dot{y}_1$ and initial conditions $y(0) = [y_1(0) \ y_2(0)]^\top = [5 \ 0]^\top$. The QoI is specified by choosing $\psi(t) = [1 \ 0]$, that is, the average value of the first component. The accumulated contributions for the Stage 1 solution using the implicit Euler method with a time step of 0.1 is shown in Figure 4.3.

The mesoscale regions are computed by first determining the global minimum of the accumulated contributions after any initial accumulation of error. The process is then repeated from the end of this mesoscale region, resulting in Algorithm 1. The mesoscale regions for the harmonic oscillator example are shown by the vertical dotted lines in Figure 4.3.

4.2. Algorithms for efficient resource allocation based on mesoscale regions. We now describe two algorithms that attempt to allocate computational resources efficiently using piecewise uniform refinement over mesoscale regions. Both approaches are based on balancing the sum of the absolute contributions of the mesoscale

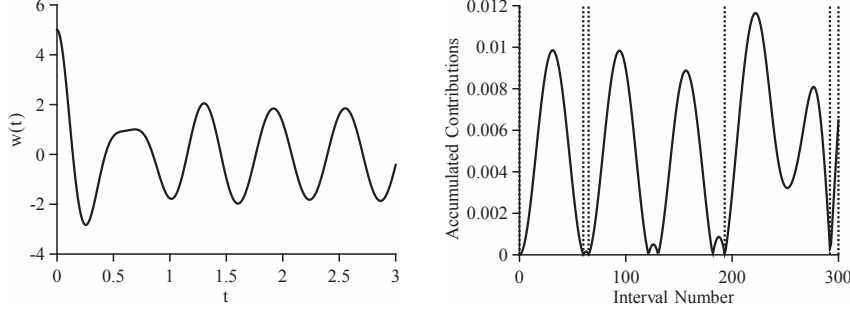


Fig. 4.3: Left: Solution of (4.7). Right: Accumulated Contributions and mesoscale regions.

Algorithm 1: Mesoscale Region Definition Algorithm

$\hat{N}$ = total number of intervals

Compute the accumulated contributions, $A\hat{E}_{\hat{k}} = \left| \sum_{i=1}^{\hat{k}} \hat{E}_i \right|$ for $\hat{k} = 1, \dots, \hat{N}$

Set $\hat{i} = 1$, $start = 1$ and $end = \hat{N}$

while $start$ is not equal to end

 Find the end of the initial increase in error $P_{\hat{i}}$ from $start$

 Find the global minimum interval $M_{\hat{i}}$ between $P_{\hat{i}}$ and end

 Set $start = M_{\hat{i}}$

 Increment $\hat{i}$

regions to the error using an argument similar to the classic Principle of Equidistribution. The algorithms attempt to solve different adaptive problems:

Allocate Fixed Resources. Determine the optimal distribution of fixed computational resources across mesoscale regions in order to minimize the error estimate.

Achieve a Given Tolerance. Determine the maximal timesteps for each mesoscale region that results in an error estimate that is smaller than a specified tolerance.

4.2.1. Allocation of fixed resources. We describe an algorithm based on deriving the optimal distribution of time steps across mesoscale regions to minimize the absolute contributions to the error from the mesoscale regions.

Let R denote the number of mesoscale regions, $\hat{N}_i$ (resp. N_i) the number of time steps for Stage 1 (resp. Stage 2) within the i^{th} mesoscale region, and $\hat{\mathcal{E}}_i$ (resp. $\mathcal{E}_i$) the error contribution for Stage 1 (resp. Stage 2) of the i^{th} mesoscale region. That is, assume that the mesoscale region i starts at interval $\hat{q}$ and ends at interval $\hat{r}$, then $\hat{\mathcal{E}}_i = A\hat{E}_{\hat{r}} - A\hat{E}_{\hat{q}}$. Note that we have $N_i = l_i h_i^{-1}$, where h_i is the length of the time steps in region i and l_i is the length of the region. We assume that the error within each mesoscale region, $\mathcal{E}_i$ satisfies,

$$\mathcal{E}_i = \frac{c_i}{N_i^{q_i}}, \quad (4.8)$$

where c_i and q_i are some constants. This implicitly assumes that a classic convergence analysis yields a bound of the error over a mesoscale region that represents the actual

asymptotic dependence on step size. The exponents q_i are determined by the order of the method. Then we have the following result by applying the minimization principle used in the classical principle of equidistribution.

THEOREM 4.1 (Fixed resources allocation). *If the number of total intervals N is fixed, $q_i = q$ for all i and the error in each mesoscale regions satisfies (4.8), then the total error is minimized if,*

$$\frac{c_i}{N_i^{q+1}} = \text{constant}, \quad \text{for all } i. \quad (4.9)$$

Proof. Since the number of total intervals is fixed, we have the constraint,

$$\sum_{i=1}^R N_i = N. \quad (4.10)$$

We minimize the total error, given by (4.8) subject to the constraint (4.10). To accomplish this we define the Lagrangian as

$$L(N_1, N_2, \dots, N_R, \lambda) = \sum_{i=1}^R \frac{c_i}{N_i^{q_i}} + \lambda \left(\sum_{i=1}^R N_i - N \right). \quad (4.11)$$

Setting the first variation of L with respect to N_i , $1 \leq i \leq R$ to 0 yields,

$$\frac{q_i c_i}{N_i^{q_i+1}} = \text{constant}. \quad (4.12)$$

Thus, the error is minimized, when (4.12) is (approximately) satisfied. Now, since $q_i = q$ for all i , we have the optimality condition,

$$\frac{c_i}{N_i^{q+1}} = \text{constant}.$$

□

During implementation, the constants c_i are approximated by using the Stage 1 error estimate (4.8) and the relation $c_i = \hat{N}_i^{q_i} \hat{\mathcal{E}}_i$, where $\hat{N}_i$ are the number of Stage 1 time steps for mesoscale region i . The above theorem provides a criteria for allocating resources for Stage 2 if the number of resources are fixed.

Numerical Example. We show the effect of applying the fixed resources allocation strategy in Theorem 4.1 for the harmonic oscillator in (4.7). Both Stage 1 and Stage 2 solutions are computed serially by employing the implicit Euler method on $[0, 3]$. The Stage 1 solution is computed with a time step of 0.01 and the error is estimated to be $-6.5E - 3$. The Stage 1 mesh has $\hat{N} = 300$ intervals. We choose $N = 3000$ for Stage 2 number of intervals. Theorem 4.1 is then applied to calculate the new time step for each mesoscale region, with the additional constraint that we do not coarsen the mesh to ensure reliability. The results using allocation strategy in Theorem 4.1 and uniform refinement (with 3000 intervals) are shown in Table 4.1. The allocation strategy in Theorem 4.1 leads to a much lower error than uniform refinement.

Refinement	Stage 2 Error
Uniform	-7.04E-04
Theorem 4.1	-1.75E-05

Table 4.1: Comparison of allocating resources using the allocation strategy in Theorem 4.1 for mesoscale regions vs. uniform refinement

4.2.2. Achieve a given tolerance. In this case, we assume that the goal is to insure that the Stage 2 error estimate is smaller than a given tolerance $TOL > 0$, i.e. so $\sum_{i=1}^R \mathcal{E}_i < TOL$. Note that by definition $\mathcal{E}_i \geq 0$. For Stage 1 errors, we assume that the total error exceeds the tolerance (otherwise, we are done, and there is no need for Stage 2). That is, we have $\sum_{i=1}^R \hat{\mathcal{E}}_i > TOL$. Let w_i be reduction factors associated with mesoscale region i , $0 < w_i \leq 1$. Then we want $\sum_{i=1}^R w_i \hat{\mathcal{E}}_i \leq TOL$. Based on this, the discretization for Stage 2 mesh is computed. For example, if the order of the method employed in both stages is q_i , then the relationship between Stage 1 and Stage 2 number of time steps for a mesoscale region i is

$$w_i = \frac{\mathcal{E}_i}{\hat{\mathcal{E}}_i} = \left(\frac{\hat{N}_i}{N_i} \right)^{q_i}, \quad (4.13)$$

from which the value of N_i , and hence the time step size of mesoscale region i , is computed.

There are many strategies to compute the factors w_i . Here we outline three strategies.

1. Choose $w_i = \frac{TOL}{\sum_{i=1}^R \hat{\mathcal{E}}_i}$, corresponding to uniform refinement.
2. Choose $w_i = \frac{TOL}{R \hat{\mathcal{E}}_i}$. This strategy, called “Mesoscale Simple”, is often (but not always) better than uniform refinement.
3. Choose w_i by solving a constrained minimization problem. This strategy, called “Mesoscale Optimized”, minimizes the objective function, $\sum_{i=1}^R N_i$, subject to the constraints that $0 < w_i \leq 1$ and $\sum_{i=1}^R w_i \hat{\mathcal{E}}_i \leq TOL$. The objective function, which is the number of intervals in the Stage 2 mesh, is formed in terms of the reduction factors w_i by employing (4.13).

Numerical Example. We show the effect of mesoscale refinement for the harmonic oscillator (4.7). Both Stage 1 and Stage 2 solutions are computed serially by employing the implicit Euler method on $[0, 3]$. The Stage 1 solution is computed with a time step of 0.01 and the error is estimated to be $-6.5E - 3$. The tolerance, TOL , for the Stage 2 solution is set to be $6.5E - 4$. The Stage 1 mesh has 300 intervals. If we use uniform refinement to meet the tolerance, we have to set the time step for Stage 2 to be 10 times smaller. The results using the two mesoscale refinement strategies, the two interval refinement strategies and uniform refinement are shown in Table 4.2. The Matlab function `fmincon` was used for solving the constrained minimization problem. The Mesoscale Simple strategy meets the refinement goal with about a third of resources compared to uniform refinement, highlighting the potential of taking the cancellation of error into account for choosing a refinement strategy. The Mesoscale Optimized strategy performs even better, achieving a lower error with the fewest number of intervals.

Refinement	Stage 2 Error	Number of intervals
Uniform	-7.04E-04	3000
Mesoscale Simple	2.17E-04	874
Mesoscale Optimized	8.76E-05	548

Table 4.2: Comparison of uniform and mesoscale refinement.

A common strategy is to (effectively) consider each interval as a mesoscale region and apply the simple and optimized strategies interval by interval. Such an approach both ignores the cancellation of error and results in a much larger optimization problem. For the same harmonic oscillator problem, the simple strategy applied interval by interval resulted in 45722 intervals and a Stage 2 error of -8.51E-06. The optimized strategy applied interval by interval resulted in 40986 intervals and a Stage 2 error of -2.15E-05. Moreover, the optimization strategy required 536572 objective function evaluations, as compared to 135 objective function evaluations for the Mesoscale Optimized strategy.

5. A posteriori analysis of the Parareal Algorithm. The Parareal Algorithm is a variation of the two stage computation. In particular, it computes corrections for the Stage 1 solution using the more accurate Stage 2 solution, and then iterates for a finite number of iterations. This is analogous to a domain decomposition approach for parallel computation. We extend the notation developed in the previous sections by the addition of a superscript (m) indicating the iteration number. Moreover, we note that in the context of the Parareal algorithm, the Stage 1 solution is often called the coarse scale solution, and the Stage 2 solution the fine scale solution [28, 2].

5.1. Alternative formulation of the Parareal Algorithm. The Parareal algorithm is given in Algorithm 2 [30, 20, 21]. The subscript p indicates values at time T_p in the algorithm. Once again, the intervals $[T_{p-1}, T_p]$ are significantly larger than the individual time steps used in the numerical integrators, $\hat{G}^p$ and F^p , for each scale.

Note that we have re-ordered the operations slightly as compared to the presentation in [28, 2], since we are primarily interested in the fine-scale solution. The evaluation of the QoI requires the discrete solutions, $Y^{(m)}$ and $\hat{Y}^{(m)}$, to be interpreted as functions of time rather than just their values at the time nodes $T_p, p = 0, \dots, P$. We obtain the fine scale solution as a function of time by considering the solutions $F^p[\hat{Y}_{p-1}^{(m)}](t)$. In this form of the Parareal algorithm, the coarse scale solutions are problematic due to the addition of the corrections $C_p^{(m)}$. In Alg. 3 we construct a Parareal algorithm which considers corrections $\bar{C}_{p-1}^{(m-1)}$ to the initial conditions for the subdomain $[T_{p-1}, T_p]$. In Lemma 5.1 we show the equivalence of our new algorithm with the original Parareal algorithm given by Alg. 2. Solutions from the new algorithm are distinguished by a superscript p , indicating they are defined over the subdomain $[T_{p-1}, T_p]$, as opposed to a subscript p which indicates solution values defined only at the times T_p .

LEMMA 5.1 (Equivalence of the Parareal algorithms). *The algorithms 2 and 3*

Algorithm 2: Parareal Algorithm

Input: y_0 : initial condition

return: $\hat{Y}_p^{(M)}$: Coarse solution at T_p , $p = 1, \dots, P$
 $Y_p^{(M)}$: Fine solution at T_p , $p = 1, \dots, P$

$C_p^{(0)} = 0$, $p = 0, \dots, P$ {initialize corrections}

for $m = 1, 2, \dots, M$

$\hat{Y}_p^{(m)} = \hat{G}^p[\hat{Y}_{p-1}^{(m)}](T_p) + C_p^{(m-1)}$, $p = 1, \dots, P$ {serial coarse solve}

$Y_p^{(m)} = F^p[\hat{Y}_{p-1}^{(m)}](T_p)$, $p = 1, \dots, P$ {parallel fine solve}

$C_p^{(m)} = Y_p^{(m)} - \hat{G}^p[\hat{Y}_{p-1}^{(m)}](T_p)$, $p = 1, \dots, P$ {update corrections}

Algorithm 3: Parareal Algorithm: Variational version

Input: y_0 : initial condition

return: $\hat{Y}^{p,(M)}(t)$: Coarse solution defined in $[T_{p-1}, T_p]$, $p = 1, \dots, P$
 $Y^{p,(M)}(t)$: Fine solution defined in $[T_{p-1}, T_p]$, $p = 1, \dots, P$

$\bar{C}_p^{(0)} = 0$, $p = 0, \dots, P$ {initialize corrections}

for $m = 1, 2, \dots, M$

$\hat{Y}^{p,(m)}(t) = \hat{G}^p[\hat{Y}^{p-1,(m)}(T_{p-1}) + \bar{C}_{p-1}^{(m-1)}](t)$, $p = 1, \dots, P$ {serial coarse solve}

$Y^{p,(m)}(t) = F^p[\hat{Y}^{p,(m)}(T_{p-1})](t)$, $p = 1, \dots, P$ {parallel fine solve}

$\bar{C}_p^{(m)} = Y^{p,(m)}(T_p) - \hat{G}^p[\hat{Y}^{p,(m)}(T_{p-1})](T_p)$, $p = 1, \dots, P$ {update corrections}

are equivalent in the sense that,

$$\begin{aligned}\hat{Y}_p^{(M)} &= \hat{Y}^{p,(M)}(T_p) + \bar{C}_p^{(M-1)}, \\ Y_p^{(M)} &= Y^{p,(M)}(T_p), \\ C_p^{(M)} &= \bar{C}_p^{(M)},\end{aligned}$$

for $p = 1, \dots, P$.

Proof.

The proof is by induction on p and m . First, the base case for the iteration. Since $C_p^{(0)} = \bar{C}_p^{(0)} = 0$, we have the equivalence for $m = 1$. Now assume that the equivalence holds for iteration $m - 1$. To show equivalence for iteration m , we perform induction on p . For $p = 1$ we have

$$\hat{Y}_1^{(m)} = \hat{G}^1[y_0] + C_1^{(m-1)},$$

whereas,

$$\hat{Y}^{1,(m)}(T_1) = \hat{G}^p[y_0 + \bar{C}_0^{(m-1)}] = \hat{G}^p[y_0],$$

and hence $\hat{Y}_1^{(m)} = \hat{Y}^{1,(m)}(T_1) + C_1^{(m-1)}$. Further, since $Y_0^{(m)} = Y^{1,(m)}(T_0) = y_0$, we have $Y_1^{(m)} = Y^{1,(m)}(T_1) = F^p[y_0]$. This also implies that $C_1^{(m)} = \bar{C}_1^{(m)}$. Now assume the hypothesis for p is true for $p-1$ such that $1 < p-1 < P$. Then,

$$\hat{Y}_p^{(m)} = \hat{G}^p[\hat{Y}_{p-1}^{(m)}] + C_p^{(m)},$$

whereas by induction hypothesis,

$$\hat{Y}^{p,(m)}(T_p) = \hat{G}^p[\hat{Y}^{p-1,(m)}(T_{p-1}) + \bar{C}_{p-1}^{(m)}] = \hat{G}^p[\hat{Y}^{p-1,(m)}(T_{p-1}) + C_{p-1}^{(m)}] = \hat{G}^p[\hat{Y}_{p-1}^{(m)}],$$

and hence $\hat{Y}_p^{(m)} = \hat{Y}^{p,(m)}(T_p) + C_p^{(m)}$. For the fine scale solutions we have,

$$Y_p^{(m)} = F^p[\hat{Y}_{p-1}^{(m)}],$$

while,

$$Y^{p,(m)}(T_p) = F^p[\hat{Y}^{p,(m)}(T_{p-1})] = F^p[\hat{Y}^{p-1,(m)}(T_{p-1}) + C_{p-1}^{(m)}] = F^p[\hat{Y}_{p-1}^{(m)}],$$

and hence $Y_p^{(m)} = Y^{p,(m)}(T_p)$. Also, since $\hat{Y}^{p,(m)}(T_{p-1}) = \hat{Y}^{p-1,(m)}(T_{p-1}) + C_{p-1}^{(m)} = \hat{Y}_{p-1}^{(m)}$, we have $C_p^{(m)} = \bar{C}_p^{(m)}$. This completes the induction for p , which in turn completes the induction for m and hence the proof. $\square$

5.2. A posteriori error analysis. We develop an error representation for the error in the QoI for the solution computed from the Parareal algorithm. The analysis can be carried out at any particular iteration of the Parareal algorithm for which error estimates are needed. Given the coarse and fine primal solutions $\hat{Y}^{p,(m)}$ and $Y^{p,(m)}$ at iteration m of the Parareal algorithm 3. Then we employ the adjoints $\hat{\phi}$, ϕ^p and $\hat{\phi}^{p,a}$ defined in (3.1), (3.8) and (3.23) respectively. This strategy ensures that the error estimates are computed in the same time complexity as that for the primal solutions $\hat{Y}^{p,(m)}$ and $Y^{p,(m)}$.

5.2.1. Coarse error representation. THEOREM 5.1.

The error in the QoI in the Stage 1 solution, $\hat{Y}^{p,(m)}$, for the Parareal algorithm 3 is given by,

$$\int_0^T (\psi, \hat{e}^{p,(m)}) dt = \sum_{p=1}^P \left[\left(-\bar{C}_{p-1}^{(m-1)}, \hat{\phi}(T_{p-1}) \right) + \hat{D}^{p,m}(\hat{Y}^{p,(m)}, \hat{\phi}) \right],$$

where $\hat{D}^p(\hat{Y}^{p,(m)}, \hat{\phi})$ is the discretization contribution term defined as

$$\hat{D}^{p,m}(\hat{Y}^p, \hat{\phi}) = \sum_{\hat{n}=1}^{\hat{N}^p} \int_{\hat{I}_{\hat{n}}^p} (\hat{\mathcal{R}}^{p,m}, \hat{\phi}) dt.$$

Here $\hat{\mathcal{R}}^{p,m}$ is the residual $\hat{\mathcal{R}}^{p,m} = -\dot{\hat{Y}}^{p,(m)} + f(\hat{Y}^{p,(m)})$.

Proof. This theorem follows directly from Theorem 3.1 by observing that $\hat{z}_{p-1} = \hat{Y}^{p,(m)}(T_{p-1}) = \hat{Y}^{p-1,(m)}(T_{p-1}) + \bar{C}_{p-1}^{(m-1)}$. $\square$

5.2.2. Fine error representation. THEOREM 5.2.

The error in the QoI in the Stage 2 solution, Y^p , for the Parareal algorithm 3 is given by,

$$\begin{aligned} \int_0^T (\psi, e^p) dt = & \sum_{p=2}^P \left[\mathcal{I}(Y^{p-1,(m)}, \hat{Y}^{p,(m)}, \hat{\phi}) + \mathcal{S}(\hat{Y}^{p,(m)}, \hat{Y}^{p-1,(m)}, \hat{\phi}, \phi^p) \right. \\ & \left. + \mathcal{A}(\mathcal{R}^{p,(m)}, \hat{\phi}^{p,a}) \right] + \sum_{p=1}^P \mathcal{D}(Y^{p,(m)}, \phi^p), \end{aligned} \quad (5.1)$$

where the four terms represent the iteration (Iter), coarse scale (Coar), auxiliary (Aux) and discretization (Disc) error contributions defined as,

$$\mathcal{I}(Y^{p-1,(m)}, \hat{Y}^{p,(m)}, \hat{\phi}) = (\hat{\phi}, Y^{p-1,(m)} - Y^{p,(m)})(T_{p-1}),$$

$$\mathcal{S}(\hat{Y}^{p,(m)}, \hat{Y}^{p-1,(m)}, \hat{\phi}, \phi^p) = (\phi^p - \hat{\phi}, \hat{Y}^{p-1,(m)} - \hat{Y}^{p,(m)})(T_{p-1}),$$

$$\mathcal{A}(\hat{\mathcal{R}}^{p,(m)}, \hat{\phi}^{p,a}) = \sum_{k=1}^{p-1} \sum_{\hat{n}=1}^{\hat{N}^k} \int_{\hat{t}_{\hat{n}-1}^k}^{\hat{t}_{\hat{n}}^k} (\hat{\mathcal{R}}^{p,(m)}, \hat{\phi}^{p,a}),$$

and

$$\mathcal{D}(Y^{p,(m)}, \phi^p) = \sum_{n=1}^{N^p} \int_{I_n^p} (\mathcal{R}^{p,(m)}, \phi^p) dt.$$

Here $\mathcal{R}^{p,(m)}$ is the residual $\mathcal{R}^{p,(m)} = -\dot{Y}^{p,(m)} + f(Y^{p,(m)})$.

Proof. The proof follows directly from Theorem 3.8. $\square$

The discretization error contribution measures the effect of choosing the Stage 2 discretization. The iteration error contribution measures the error due to insufficient convergence of the algorithm whereas the coarse scale error contribution involves the difference of fine and coarse adjoint solutions scaled by coarse scale initialization differences at the end of subdomains.

5.2.3. Numerical examples. We consider two numerical examples.

Harmonic Oscillator. We show the accuracy of the estimate for the Parareal algorithm for the harmonic oscillator (4.7). The QoI is again specified by choosing $\psi(t) = [1 \ 0]$, that is, the weighted average value of the first component. The numerical solvers, $\hat{G}^p$ and F^p are chosen to be implicit Euler on the Stage 1 and Stage 2 time meshes on subdomain p respectively. We choose $P = 10$ and $T = 3.0$. We choose time step for Stage 1 to be 0.01 and compute the initial coarse scale error as $-6.5E-3$. We set an error tolerance of $-6.5E-4$ and employ uniform refinement and the Mesoscale Optimize refinement in § 4.2.2 to construct the Stage 2 discretization.

The uniformly refined mesh for a Stage 2 has time step of 0.001 for a total of 3000 intervals. We show the results for the error and its different contributions for Stage 2 in Table 5.1 for the first four iterations of the Parareal algorithm. The error decreases rapidly initially, but plateaus around the fourth iteration. This is explained by examining the different error contributions. Initially the iteration contribution dominates, as might be expected. By iteration 4, the iteration error is an order of

m	Error Est.	Effec. Rat.	Disc	Iter	Coar	Aux
1	0.0135	1.00	-7.65E-04	0.0142	0	1.25E-06
2	-0.0024	1.00	-6.68E-04	-0.0017	-2.05E-06	1.96E-06
3	-8.85E-04	1.00	-6.87E-04	-1.97E-04	-1.79E-06	2.05E-06
4	-7.01E-04	0.90	-7.01E-04	-7.13E-05	-1.81E-06	2.00E-06

Table 5.1: Error contributions for Parareal algorithm for a uniformly refined Stage 2 mesh. Here Error Est. is given in (5.1), Disc, Iter, Coar and Aux are the various contributions to the error by discretization, iteration, coarse scale and auxiliary terms, see (5.1) and §3.3.

magnitude smaller than the discretization error. Further iterations do not have a significant impact on decreasing the error.

The Stage 2 mesh obtained using the Mesoscale Optimize refinement has 549 intervals which is more than 5 times less than the corresponding number for uniform refinement. The results are shown in Table 5.2. The results are qualitatively similar to those in Table 5.1.

m	Error Est.	Effec. Rat.	Disc	Iter	Coar	Aux
1	-0.0034	1.00	-3.75E-04	-3.0E-03	0	9.40E-07
2	4.29E-04	0.99	5.01E-05	3.79E-04	-1.09E-06	8.61E-07
3	1.034E-04	0.97	8.42E-05	1.93E-05	-1.06E-06	8.75E-07
4	8.44E-05	0.96	8.46E-05	3.39E-09	-1.05E-06	8.75E-07

Table 5.2: Error contributions for Parareal algorithm for a Stage 2 mesh obtained by using Mesoscale Optimize refinement. Here Error Est. is given in (5.1), Disc, Iter, Coar and Aux are the various contributions to the error by discretization, iteration, coarse scale and auxiliary terms, see (5.1) and §3.3.

A nonlinear partial differential equation. Consider

$$\begin{cases} \dot{u} + \frac{1}{2} \cos(2\pi t)(1+u)u_x = 0, & (x, t) \in [0, 1] \times (0, T], \\ u(x, 0) = \sin(2\pi x), & x \in [0, 1], \end{cases}$$

with periodic boundary conditions. The QoI is the integral $\int_0^T \int_{0.5}^{0.6} u(x, t) dx dt$. A first order finite difference scheme with spatial discretization parameter $h = 0.05$ leads to a system of twenty ODEs. We set $T = 2.0$, $P = 10$ and choose time step for Stage 1 to be 0.001 for a total of 2000 time intervals. The initial coarse scale error was $-3.98E-04$. We set an error tolerance of $-3.98E-05$ and compute a Stage 2 mesh by employing uniform refinement and the Mesoscale Optimization refinement.

The Stage 2 mesh obtained by the uniform refinement has 20000 intervals. The results of four iterations of the Parareal algorithm are shown in Table 5.3. The iteration error contribution is significantly less than the discretization contribution

m	Error Est.	Effec. Rat.	Disc	Iter	Coar	Aux
1	-3.89E-04	1.00	-3.97E-05	-3.50E-04	0	5.42E-07
2	-4.16E-05	1.00	-3.99E-05	-1.75E-06	-1.31E-07	1.95E-07
3	-3.98E-05	1.00	-3.99E-05	8.94E-09	-1.35E-07	1.91E-07
4	-3.99E-05	1.00	-3.99E-05	1.10E-10	-1.35E-07	1.91E-07

Table 5.3: Error contributions for Parareal algorithm for a Stage 2 mesh obtained by using uniform refinement. Here Error Est. is given in (5.1), Disc, Iter, Coar and Aux are the various contributions to the error by discretization, iteration, coarse scale and auxiliary terms, see (5.1) and §3.3.

at the end of iteration 2, indicating that the algorithm may be stopped at this point since further iterations will not decrease the error.

The Stage 2 mesh obtained by the Mesoscale Optimization refinement has 13935 intervals, which is again significantly less than uniform refinement. The results of four iterations of the Parareal algorithm are shown in Table 5.4.

m	Error Est.	Effec. Rat.	Disc	Iter	Coar	Aux
1	-3.72E-04	0.99	-3.77E-05	-3.35E-04	0	7.85E-07
2	-3.76E-05	0.99	-3.78E-05	2.56E-07	-3.06E-07	2.89E-07
3	-3.78E-05	0.99	-3.78E-05	4.39E-09	-3.07E-07	2.89E-07
4	-3.78E-05	0.99	-3.78E-05	9.28E-13	-3.07E-07	2.89E-07

Table 5.4: Error contributions for Parareal algorithm for a Stage 2 mesh obtained by using Mesoscale Optimize refinement. Here Error Est. is given in (5.1), Disc, Iter, Coar and Aux are the various contributions to the error by discretization, iteration, coarse scale and auxiliary terms, see (5.1) and §3.3.

6. Conclusions. We derive a number of a posteriori error estimates for a quantity of interest for a general class of numerical methods for solving an initial value problem that employ a two stage computation consisting of a coarse and a fine solve. The various estimates utilize different adjoint problems that present trade-offs between accuracy, reliability, and computational cost. We present two applications of the error estimates. In the first, we construct two algorithms for the distribution of computational resources that exploit cancellation of error that occurs over time regions. In the second, we extend the analysis to the Parareal Algorithm for parallel-in-time solution, including the effects arising from finite iteration in the analysis.

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